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Real-Time American Sign Language Alphanumeric Detection Using a Convolutional Neural Network

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Abstract

This project develops a real-time ASL detector that identifies 36 distinct hand signs for the digits 1–10 and the letters A–Z directly from a laptop webcam.

A YOLOv11 model was fine-tuned on a Roboflow-hosted dataset of 19,365 annotated images and integrated into a standalone Python application using OpenCV and Tkinter. The system runs entirely on consumer hardware with no cloud services or paid APIs. This is mean to show that open-source detection models can deliver accessible, locally-deployable sign-language recognition.

Methods and Materials

Dataset: 19,365 Roboflow-annotated images with 36 classes, covering ASL signs for 1-10 and A-Z in YOLO bounding-box format.
Architecture: YOLOv11 nano (~2.6M parameters), pretrained on Microsoft COCO and fine-tuned on the ASL dataset.
Training: 20 epochs at 640×640, batch; trained on a Google Colab T4 GPU in ~2.5 hours.
Deployment: Python application using OpenCV for video capture and Tkinter for the user interface.

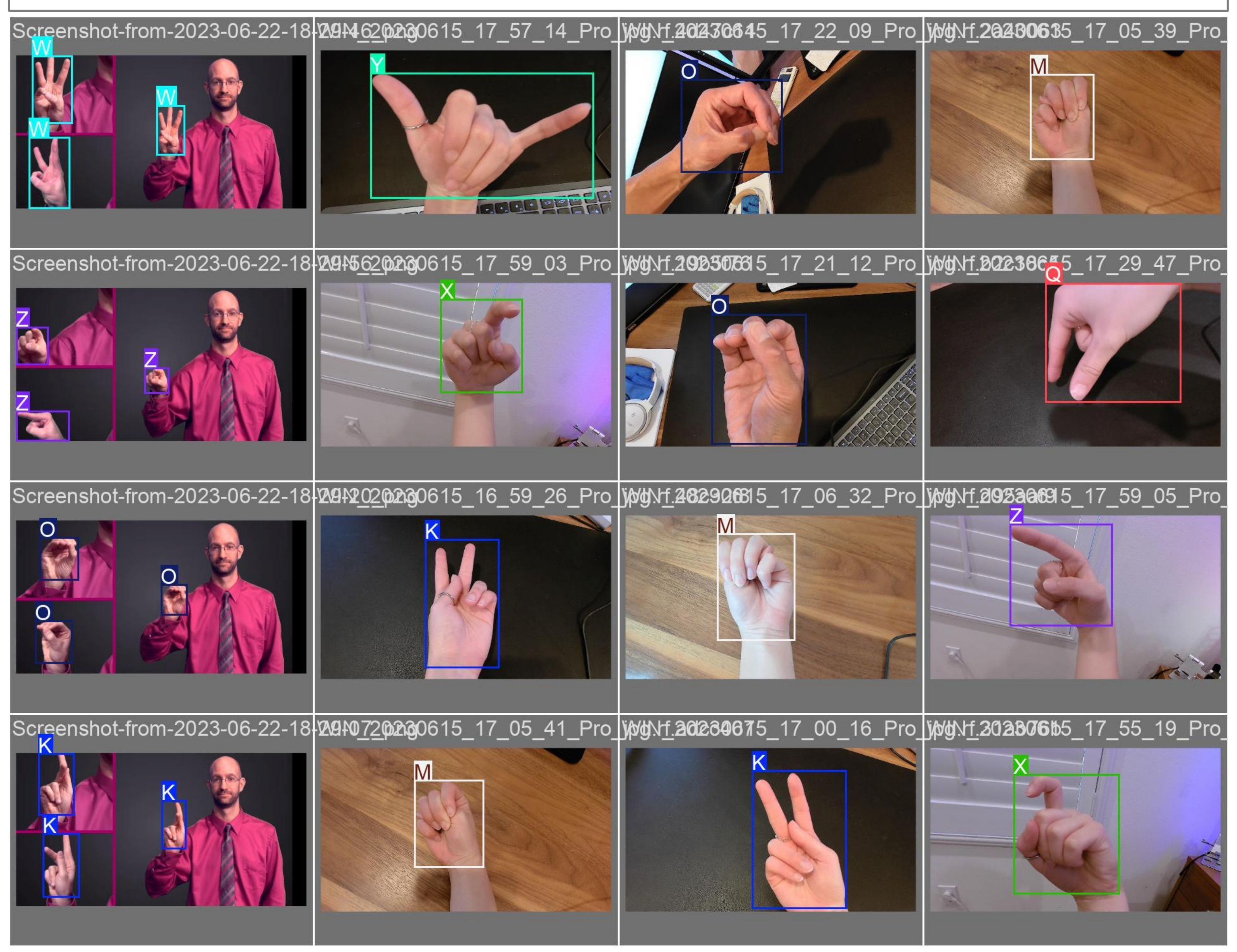

Figure 1. Sample from Dataset.

Results

The model achieved **96.4% average per-class accuracy** across the 26 letters, with B and Y reaching 100% accuracy and P being the worst-performing letter at 88.9%.

The digits 1–10 were underrepresented in the training data and produced few reliable detections, leaving the model most effective on the letter classes.

The most common confusions involved **P↔Q (9%)** and **T↔S/N (5% each)**, pairs that share visually similar handshapes.

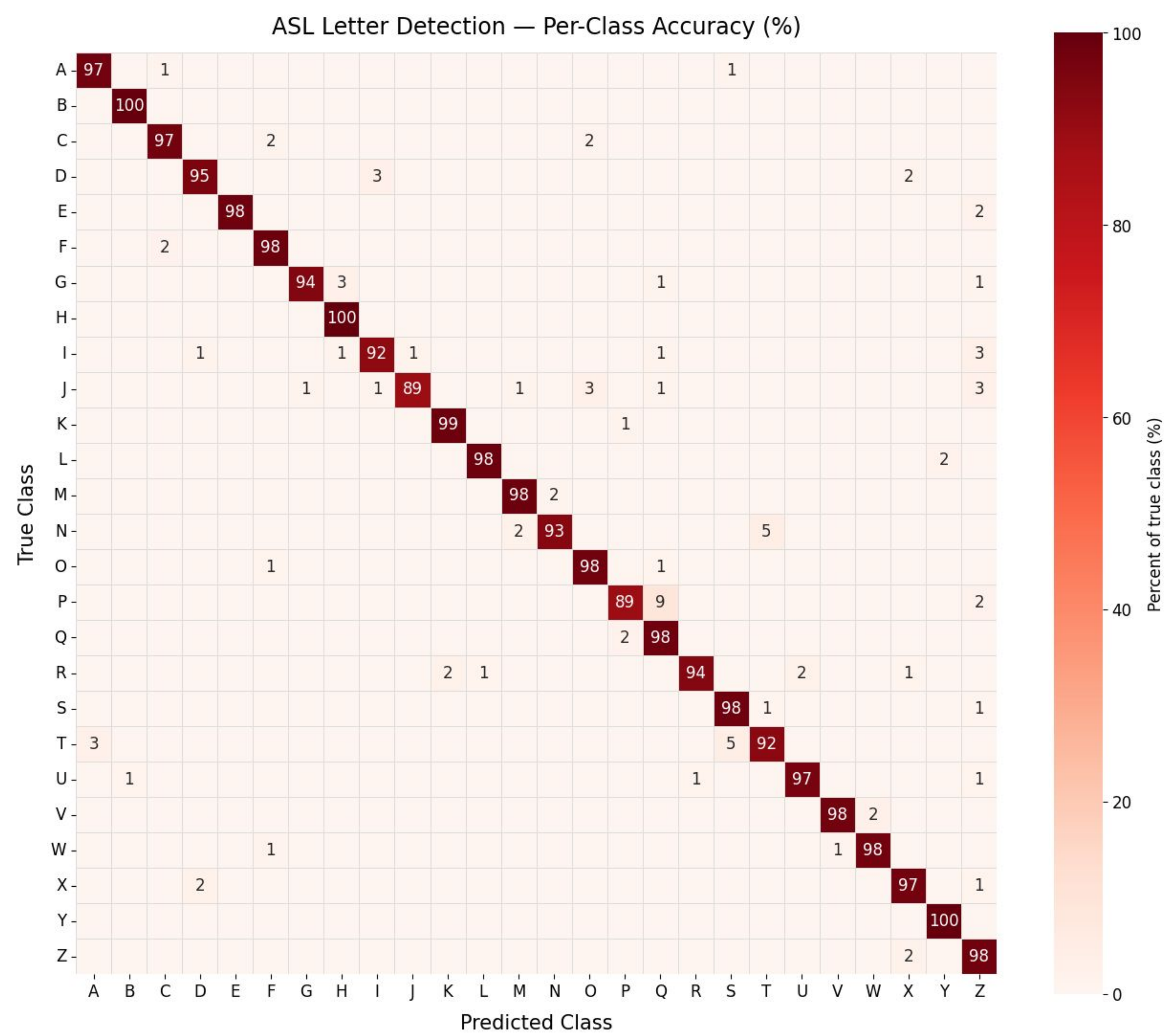

Chart 1. Confusion Matrix of A-Z Class

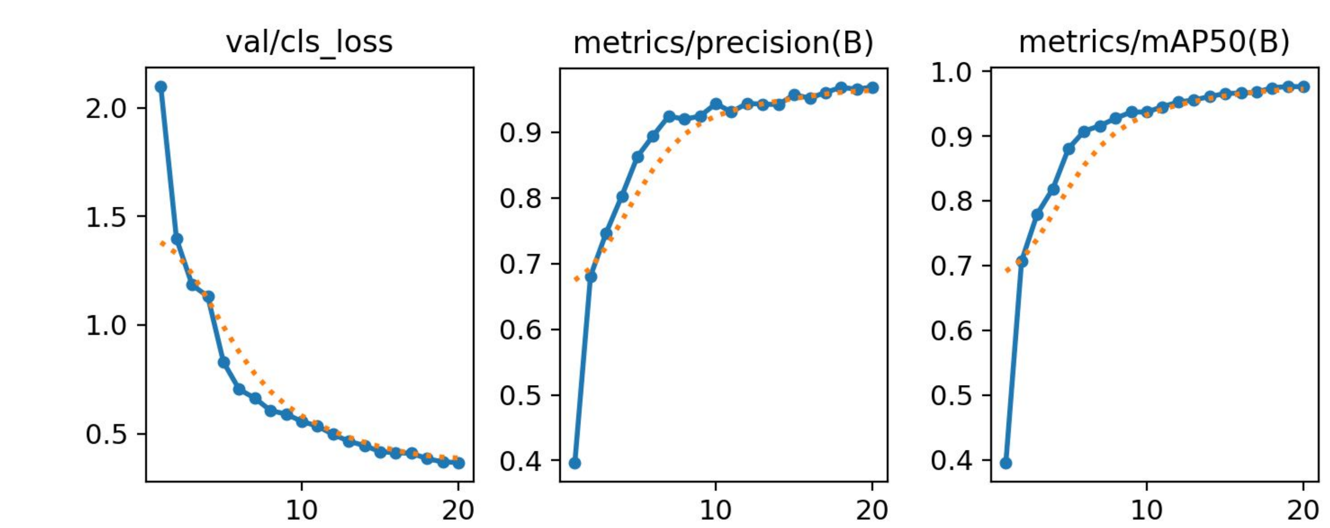

Chart 2. Left to Right: Validation Loss curve; Precision over Training curve; mAP@IoU0.5

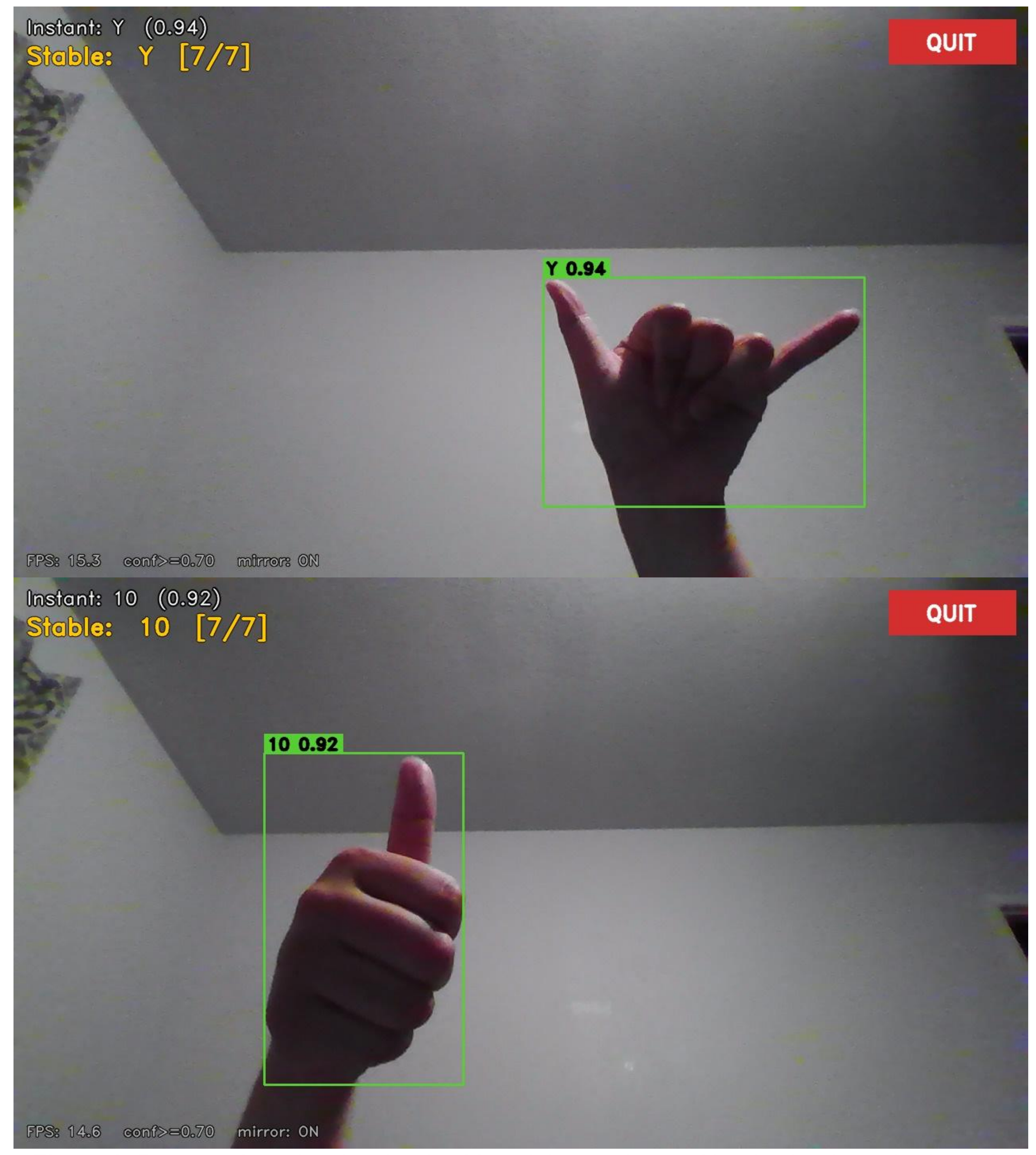

Figure 2. Program detecting the sign for "Y" and the starting position for the dynamic sign "10"

Discussion

The digits 1–10 were underrepresented in the dataset and were rarely detected in the validation process. More data of the numeric signs would fix this issue.
Dynamic signs which required motion,10, J, and Z, were learned only as their starting positions. Research into a model which assesses frame changes over time could solve this issue. It was not a priority as this model would only use it for 3 of 36 signs.

Conclusions

A real-time ASL sign detection system was successfully implemented and runs end-to-end on standard laptop hardware, achieving strong accuracy on the static letters A–Z. Future work includes rebalancing the training set to give the digits 1–10 representation equal to the letters, adding support so that motion-based signs are recognized by their full gesture rather than their starting frame.

ultralytics YOLO 11 · roboflow · python · OpenCV

References

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Datasets

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2. https://universe.roboflow.com/putsanun/asl-language-gvkgn
3. https://universe.roboflow.com/compvi/american-sign-language-letters-extend
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